CrCD: Multidirection-MLP-Based Cross-Contrastive Disambiguation for Hyperspectral Image Partial Label Learning

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***Abstract*— Due to the intricate nature of the hyperspectral image (HSI) and the constraints imposed by annotators' limited prior knowledge, the collection of HSI data poses significant challenges. Although there has been growing attention toward the issue of transfer learning and few-shot learning, partial label learning (PLL) has received scant attention in HSI classification. PLL refers to a scenario where training samples are associated with a set of candidate labels, among which only one is the correct label. Consequently, PLL holds significant practical importance for HSI classification, as it can alleviate the costs associated with HSI labeling. In this article, a cross-contrastive disambiguation (CrCD) method is proposed based on multidirection multilayer perceptrons (MLPs) for HSI PLL, which includes two main components: representation learning for feature extraction and label disambiguation strategy for PLL. First, we design a heterogeneous network framework composed of 2-D encoder and 3-D encoder, and introduce a multidirection MLP and a multiscale attention for long-range spatial–spectral information. Second, apart from enforcing consistency in feature representation, we pioneer the establishment of a consistency constraint on semantic prediction probabilities with contrastive learning. Furthermore, the cross-label disambiguation strategy is introduced to provide reliable guidance for network training. Extensive experiments demonstrate that CrCD outperforms several current state-of-the-art (SOTA) approaches in HSI PLL and achieves results comparable to fully supervised learning. Code: https://github.com/Nemo96yu/CrCD.**

Received 16 June 2024; revised 27 August 2024; accepted 11 October 2024. Date of publication 21 October 2024; date of current version 4 November 2024. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant 2024CDJYXTD-009; in part by the National Natural Science Foundation of China under Grant 62371076, Grant 62071340, and Grant 62201109; and in part by the Natural Science Foundation of Chongqing under Grant CSTB2022NSCQ-MSX0452. *(Corresponding author: Fulin Luo.)*

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Digital Object Identifier 10.1109/TGRS.2024.3483989

***Index Terms*— Hyperspectral image (HSI) classification, label disambiguation, partial label learning (PLL), representation learning.**

I. Introduction

With the rapid advancement of remote sensing imaging techniques, a mount of hyperspectral images (HSIs) can now be acquired. Renowned for its wealth of spatial and spectral information about objects, HSIs are extensively employed in various fields such as environmental surveillance, military reconnaissance, and deep space exploration [1], [2]. As a vital foundation for these applications, HSI classification entails assigning a distinct label to every pixel and has attracted extensive attention from both academia and industry.

In the early research on HSI classification, a mount of methods focused on exploring the role of HSI spectral features, such as support vector machine (SVM) [3] and multinomial logistic regression [4]. However, due to the neglect of spatial context, the resultant classification maps from these pixel-level classifiers were found to be suboptimal. Subsequently, a large number of advanced techniques, such as extended morphological profiles (EMPs) [5], and composite kernels and morphological kernels for multikernel learning [6], aim at probing the spectral–spatial intricacies. Most recently, the advent of deep learning technology has engendered novel research paradigms for HSI processing and there have been three types of primary deep learning frameworks for HSI classification tasks, such as convolution neural network (CNN) [7], [8], attention networks [9], [10], [11], and multilayer perceptron (MLP) [12], [13]. Guo et al. [14] proposed a CNN-enhanced multilevel Haar wavelet features fusion network (CNN-MHWF2N) to obtain sufficient spectral–spatial features. Chen et al. [15] proposed a model called grid network (GNet) based on CNN to represent spectral and spatial features. CNNs have a clear advantage in capturing local semantic information, but they are not effective in capturing global information [16]. To overcome the above shortcoming, Sun et al. [9] proposed a spectral–spatial feature tokenization transformer (SSFTT) method to capture spectral–spatial features and high-level semantic features, which can capture

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long-range dependency information. Sun et al. [12] proposed a spectral-spatial MLP network with reciprocal points learning (SSMLP-RPL) to improve the performance of HSI classification. Although the powerful feature extraction capabilities of deep learning methods make the information from HSI more accurate and achieve the satisfactory classification results, they require a substantial amount of accurately annotated data as supervised information to learn the intrinsic relationship between pixel feature information and pixel categories. Actually, due to the necessity of annotating the pixel in HSIs and the limitations of annotators' prior knowledge, the collection of training samples is expensive and time-consuming.

To alleviate the cost and burden of annotation, researchers are exploring the learning paradigms with few labeled training samples for HSI classification, such as transfer learning [17] and few-shot learning [18]. Transfer learning first trains the model on a large-scale source domain, and then transfers the model to a small-scale target domain to learn the distribution of new data [19]. Yang et al. [20] introduced to leverage transfer learning to train the deep model for HSI classification when the number of training samples is limited. Zhou et al. [17] advocated for combing spectral–spatial feature extraction with metatransfer learning to address HSI classification with insufficient labeled samples. Few-shot learning has been proposed for HSI classification to train the model with only a few labeled samples [21]. Wang et al. [22] proposed the heterogeneous few-shot learning to train the HSI classification model with a few labeled samples per class. Xiao et al. [23] introduced a neural architecture search-based few-shot learning for HSI classification. In summary, both transfer learning and few-shot learning just reduce the reliance on the number of label samples without taking the inherent label ambiguity into consideration, which is the intrinsic factor leading to the challenge of labeling pixels in the HSI. Specifically, in the processing of HSI annotation under real-world scenarios, the complexity of land covers makes it difficult for annotator to determine which class labels the pixels belong to [24]. As shown in Fig 1, there is a confusing area with low interclass dissimilarity, thus several class labels all seem to be the pixel's ground-truth label in this area and consist of the candidate label set. In such a case, partial label learning (PLL) is a common framework that treats the candidate label set as the coarse labeling information to train the model.

PLL refers to a classic task in weakly supervised learning, allowing each training sample to be associated with a candidate label set that includes the ground-truth label, which was proposed to solve the label ambiguity in many real-world annotation scenarios of natural images [25]. The core challenge in PLL is label disambiguation, i.e., identifying the ground-truth label from the candidate label set [26]. The existing PLL methods can be divided into two main categories: average-based methods and identification-based methods according to label disambiguation. The average-based methods [25], [27] treat all the candidate labels equally and then average the model's outputs for all candidate labels to obtain the final prediction. Nonetheless, the main and apparent limitation is that the false positive labels can significantly mislead the predictions. The identification-based

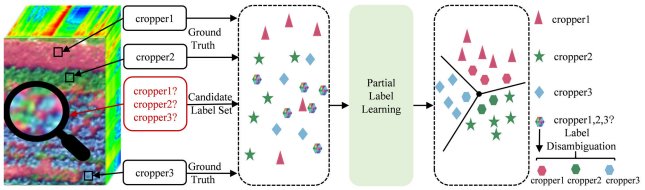


Fig. 1. Flowchart of PLL. The core step is to select the ground-truth label from candidate label set during the training process, called label disambiguation strategy.

methods treat the true labels as latent variables and optimizes them iteratively [28], [29], [30]. However, the identified labels may not be the ground-truth label in the candidate set. Recently, researchers make efforts to improve the quality of representation learning for reliability of label disambiguation. Wang et al. [26] proposed a contrastive label disambiguation model (PiCO), which consists of a contrastive learning module along with a novel class prototype-based label disambiguation algorithm. He et al. [31] proposed a weighted calibration rank loss to constrain the representation learning process. Xia et al. [32] presented a guided prototypical classifier (PaPi) for effective visual representation learning. However, these methods were designed for PLL on natural image and might not represent satisfactory performance on HSI PLL. There is very little work [24], [33] to design PLL method for HSI, which refers to allowing the training pixel with a candidate label set. Yang et al. [24] deems that PLL should be emphasized in HSI classification and proposed a superpixelwise low-rank approximation (LRA)-based PLL method (SLAP), which makes them the pioneers in applying PLL in HSI classification.

In this article, we further analyze the challenge in PLL for HSI classification task. On the one hand, under the limited supervised information, how to effectively extract features from HSI is more difficult than natural image due to the spectral variation, i.e., the spectrum from the same category may change greatly. In such a case, the spatial information in HSI is usually combined to capture discriminative features of different categories. However, CNN limited by the kernel size is poor at capturing global information, and the transformer-based self-attention mechanism [34], [35] that is good at extracting global information is computationally complex [16]. Recent studies have demonstrated the computational friendliness of simple MLP on long-rang feature extraction [12], [36]. However, MLP can only capture long-range information at the single-dimensional direction and cannot adequately represent the spatial–spectral information in HSIs. Therefore, improving MLP to handle long-range information at different directional dimensions may be a good strategy, which can be used to fully exploit the long-rang information of HSI to provide the reliable semantic representation for classification. On the other hand, label disambiguation usually requires the effective feature representations and plays a very important role in representation learning. Nonetheless, the dependence on representation learning has given rise to a significant predicament: the inherent ambiguity in labeling can unfortunately become apparent in the process of representation learning, potentially impeding the

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quality of feature representation, and consequently hindering the effect of label disambiguation.

To achieve the virtuous circle between representation learning and label disambiguation on HSI, we introduce a coherent and synergistic framework called multidirection-MLP-based cross-contrastive disambiguation (CrCD) for HSI PLL, which can capture discriminative information of land covers to assist label disambiguation. Specifically, our framework consists of two key components. First, we construct two heterogeneous networks at 3-D and 2-D respectively based on contrastive learning to fully capture discriminative features from HSI data for classification, which is unexplored in previous HSI classification literatures. In this part, we take advantage of the convolution layer to extract the shallow spectral and spatial features. To compensate for the fact that MLP can only capture information in single dimension, we design a multidirection MLP layer and a multiscale attention mechanism to capture the rich long-range dependencies information. To leverage the complementarity of features and achieve feature consistency, we construct a consistency constraint loss on both features and semantics. Second, we develop a novel cross-label disambiguation strategy based on the learned feature representation, which can partially alleviate the direct entanglement between representation learning and label disambiguation. By interchanging the above two steps, CrCD converges toward a solution with a highly distinguishable representation for precise classification. Empirically, CrCD achieves the state-of-the-art (SOTA) performance on three benchmark datasets, surpassing the recent advanced methods by a considerable margin and achieving the results that are on par with a few fully supervised learning methods. Our main contributions are summarized as follows.

1) Two heterogeneous networks (2-D encoder and 3-D encoder) based on contrastive learning are proposed, and the multidirection MLP and the multiscale attention are introduced. The multidirection MLP is constructed to comprehensively capture long-distance spatial information in different direction, and the multiscale attention is designed to extract the global spatial information. In this way, we can fully capture the long-range spatial and spectral information for high-quality representation and provide more reliable information for classifier.
2) Apart from only establishing consistency constraints on the feature representation of encoders as the previous work, we first establish consistency constraints on semantic prediction probabilities of classifiers, which in turn achieves the complementarity between the classifiers and thus mitigates the negative impact of the ambiguous feature representation on the classifiers. In this way, we develop a new perspective on consistency constraints to better achieve the complementary advantages between the two heterogeneous networks.
3) A novel cross-label disambiguation strategy is proposed for reliable guidance on network training, namely exchanging the prototype prediction results of two different networks and using them in their respective label disambiguation processes. In this way, it is possible to mitigate the entanglement of representation learning and label disambiguation in a single network.

The rest of this article is organized as follows. The problem of PLL on HSI is defined in Section II. The proposed CrCD framework is described in Section III. Section IV conducts some experiments on three real HSI datasets to demonstrate the effectiveness of CrCD. In Section V, we analyze some conclusions of this article and suggestions for future work.

II. Preliminaries

In this section, we introduce some symbols to define the problem of HSI PLL. Given the original HSI data $\mathcal{X} \in \mathbf{R}^{h\times w\times b}$, where $h\times w$ is the spatial size and b is the number of spectral bands. Let $\mathcal{D} = \{x_i, Y_i\}_{i=1}^n$ be the partially labeled training dataset with n pixels. Each pixel $x_i \in \mathcal{X}$ has b spectral dimension and corresponds to a candidate label set $Y_i \in \mathcal{Y}$, where $\mathcal{Y} \in \{1, 2, \ldots, K\}$ denotes the label space. Unlike the setting of traditional supervised HSI classification that the ground-truth label y_i is definitely known, the ground-truth label for PLL is concealed within a candidate label set. The objective of HSI PLL is to learn a multiclass classifier with the capacity to accurately predict the true label of the unseen pixel. It is obvious that directly utilizing the partially labeled dataset $\mathcal{D}$ to train a classifier would be susceptible to the misguidance from label ambiguity in Y_i. Thus, there is an essential label disambiguation process associated with PLL methods, which means to identify the ground-truth label y_i from the candidate label set Y_i.

For the traditional PLL, there are two main steps to obtain the classifier. Label disambiguation is the first step that aims to transform the partial-labeled training dataset into the common supervised form through selecting the latent true label. And then, the second step is to train the classifier with the disambiguated labels. For deep learning-based PLL in an end-to-end training manner, it needs to train the deep neural network (including the feature extractor $\mathcal{X} \to \mathbb{R}^d$ and the classifier $\mathbb{R}^d \to [0, 1]^K$) and perform the label disambiguation simultaneously. In this case, each pixel x_i is generally associated with a label confidence vector $p \in [0, 1]^K$, where each entry means the probability of the corresponding label being the ground truth. The process of label disambiguation is to update the label confidence and provides supervised information for the training of feature extractor. In this way, the weight of ground truth will gradually increase to 1, conversely, decrease to 0. Concurrently, p is treated as coarse supervision to guide the training process of networks.

In our framework CrCD, for each pixel, we take the advantages of patch-based sampling strategy to cut the original data into cubes ($x_i \in \mathbf{R}^{s\times s\times b}$) for each pixel and make use of principal component analysis (PCA) to reduce the spectral dimension from b to l for generating the augmented data cubes ($x_i' \in \mathbf{R}^{s\times s\times l}$), which is treated as two views and obtains two types of input data for network training ($\mathcal{D}_1 = \{x_i', Y_i\}_{i=1}^n$ and $\mathcal{D}_2 = \{x_i, Y_i\}_{i=1}^n$). In this way, we can make full use of spatial and spectral information in HSI and enhance the efficacy of feature extraction.

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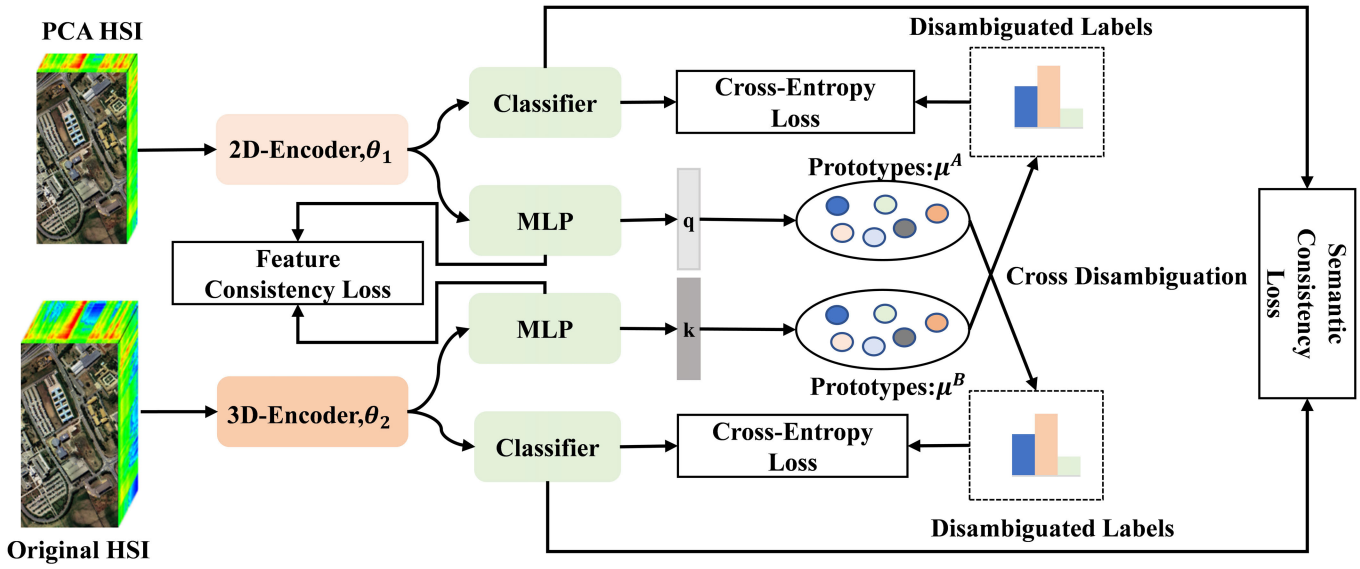


Fig. 2. Flowchart of the proposed CrCD framework.

III. Method

In this section, we introduce our novel multidirection-MLP-based CrCD framework for HSI PLL in detail. As shown in Fig. 2, CrCD consists of two essential components to enhance representation quality (Section III-A) and reduce label ambiguity respectively (Section III-B), which systematically works as a whole and mutually influences each other.

As mentioned above, a high-quality representation is crucial to foster a virtuous cycle between representation learning and label disambiguation in PLL. In CrCD, we attain this objective by constructing two heterogeneous networks to learn spectral–spatial feature representation, which is treated as the foundation of label disambiguation. Specifically, we design a 2-D-Encoder and a 3-D-Encoder for capturing spatial–spectral information in 2-D space and in 3-D space, respectively. And then we achieve consistency and complementarity between them based on contrastive learning.

A. Heterogeneous Networks for HSI PLL

1) 2-D-Encoder: As shown in Fig. 3, we primarily explore the spatial information from the data cubes processed by PCA (x_i', Y_i). First, we obtain the local spatial information based on three 2-D convolution layers in series, where the inputs of the later convolution layer are the outputs of the previous convolution layer, except for the first convolution layer whose input is x_i'. Taking the first layer as an example, the processing can be expressed as follows:

$$F_i' = f_{\mathrm{GELU}}\left(f_{BN}\left(f_{\mathrm{conv}}^{k\times k}\left(x_i'\right)\right)\right) \tag{1}$$

where $f_{\mathrm{conv}}^{k\times k}(\cdot)$ denotes the 2-D convolution function with the kernel size of $k \times k$, and $f_{\mathrm{GELU}}(\cdot)$ and $f_{BN}(\cdot)$ are the Gaussian error linear unit and batch normalize function, respectively. In this way, we can obtain feature representation F_i' with the size of $s \times s \times d$, where s is the size of height and width, and d is the feature dimension size.

After the three 2-D convolution layers, we can initially obtain local spatial information F_i', which is not enough to provide a high-quality representation for label disambiguation. Aiming at the long-distance spatial information in the data cubes, we make use of the MLP layer to model the spatial correlation of long distance at three directions called multidirection MLP. Specially, we first split the data cubes F_i' along the vertical, horizontal, and channel directions respectively and obtain three formats of data ($F_i'^{v} \in R^{s\times sd}$, $F_i'^{h} \in R^{s\times sd}$, and $F_i'^{d} \in R^{ss\times d}$). And then, we take these formats to pass the MLP layers with different weights, which can be expressed as follows:

$$F_i'^{v} = f_{\mathrm{mlp_}v}\left(F_i'^{v}\right) \tag{2}$$

$$F_i'^{h} = f_{\mathrm{mlp_}h}\left(F_i'^{h}\right) \tag{3}$$

$$F_i'^{d} = f_{\mathrm{mlp_}d}\left(F_i'^{d}\right) \tag{4}$$

where $f_{\mathrm{mlp}_c}(\cdot)$, $f_{\mathrm{mlp}_c}(\cdot)$, and $f_{\mathrm{mlp}_c}(\cdot)$ denote the MLP function along the vertical, horizontal, and channel directions, respectively [36]. And then, we reshape $F_i'^{v}$, $F_i'^{h}$, and $F_i'^{d}$ into the original shape and obtain fused output $\hat{F}_i'$ by the recalibration-weighted fusion approach proposed in [36]. Finally, we feed $\hat{F}_i'$ into a fully connected layer and add it to F_i' as follows:

$$F_i' = F_i' + f_{fc}\left(\hat{F}_i'\right) \tag{5}$$

where $f_{fc}(\cdot)$ denotes the fully connected function.

In this way, we can jointly model spatial semantics at long distance. Moreover, to overcome the limitations of FC in processing 2-D spatial information, we design a multiscale attention network based on 2-D convolution layers to complement the global spatial information in 2-D space. Specially, it is made up of the convolution layers with different receptive fields: three spatial convolutions with the same kernel size and different dilation rate, and a convolution layer with kernel size 1, which is used to capture the spatial attention maps at

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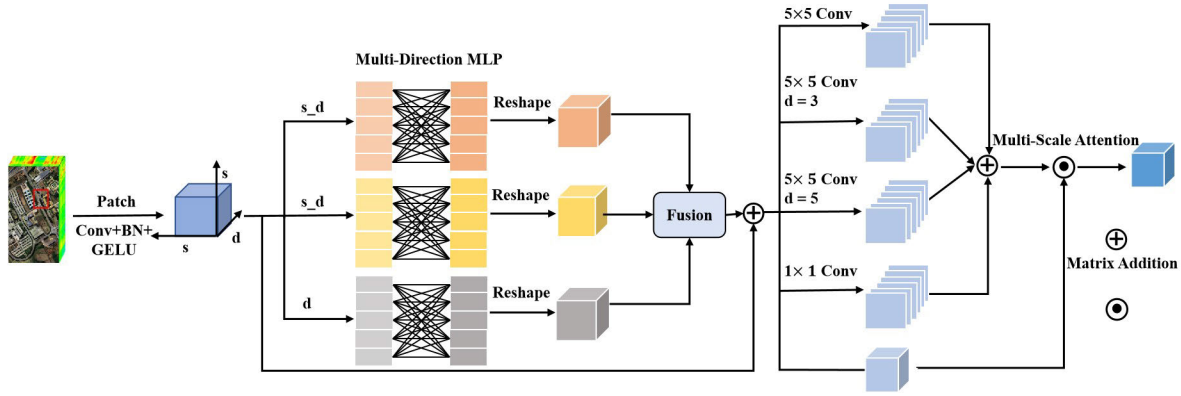


Fig. 3. Structure of the 2-D-Encoder.

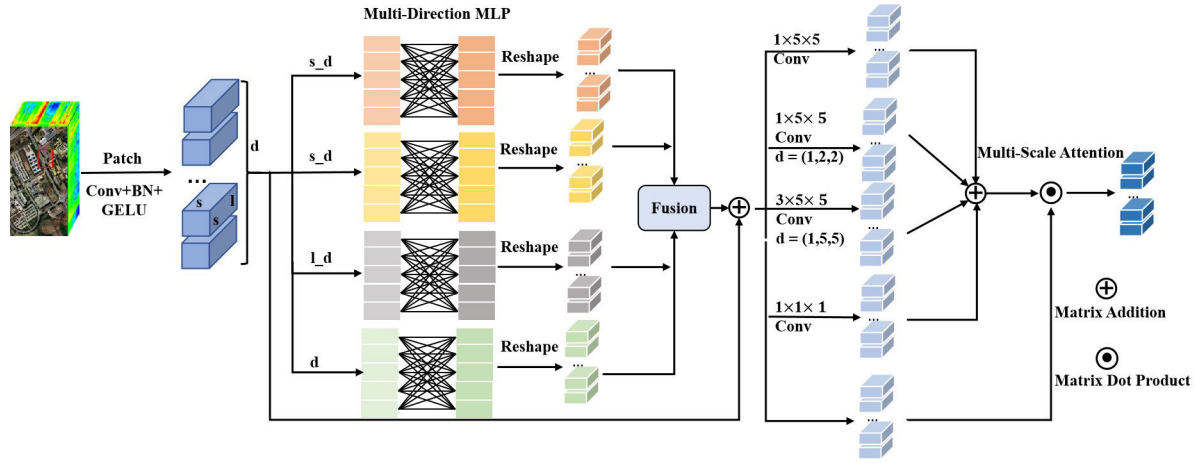


Fig. 4. Structure of the 3-D-Encoder.

varying scales that overrides the global range at 2-D space. We take F_i' from (5) as the input and generate the attention map as follows:

$$S' = f_{\text{conv}}^{1\times1}(F_i') + \sum_{r}^{\{1,2,5\}} f_{\text{conv}(r)}^{5\times5}(F_i') \tag{6}$$

where $f_{\text{conv}(r)}^{k\times k}(\cdot)$ denotes the 2-D convolution function with the dilation rate of r on a kernel size of k. Specifically, we obtain the attention map S at global range by adding all of the outputs from the convolution layers with different convolution range, and multiply them with the input features for obtaining the output features, which is different from the calculation process of the traditional attention method [37]

$$F_i' = F_i' + S' \otimes F_i'. \tag{7}$$

In this way, we can enhance the spatial information in 2-D space efficiently compared with the traditional ones.

2) 3-D-Encoder: As shown in Fig. 4, to compensate for the spectral information loss in the process of 2-D-Encoder that mainly mines spatial information, we take the original data cubes (x_i, Y_i) as input to capture the spectral–spatial information simultaneously, which is a little different from 2-D-Encoder. First, we explore the spectral–spatial features through three 3-D convolution layers in series, and the first layer can be expressed as follows:

$$F_i = f_{\text{GELU}}\left(f_{BN}\left(f_{\text{conv}}^{m\times k\times k}(x_i)\right)\right) \tag{8}$$

where $f^{m\times k\times k}(\cdot)$ denotes the 3-D convolution function with the kernel size of $m \times k \times k$, and the next layer's input is its previous layer's output. After that, we can obtain the feature representation F_i with the size of $l \times s \times s \times d$, where s is the size of height and width, l is the spectral size, and d is the feature dimension size.

In this way, the spectral–spatial information within the local space can be explored effectively, which is expressed as the data cubes F_i with the dimension number d. To further explore the correlation information at long distance, we take the advantages of MLP layers at four directions to model contextual spatial information and long-distance spectral features, which can be expressed as follows:

$$F_i^v = f_{\text{mlp_}v}(F_i^v) \tag{9}$$
$$F_i^h = f_{\text{mlp_}h}(F_i^h) \tag{10}$$
$$F_i^l = f_{\text{mlp_}l}(F_i^l) \tag{11}$$
$$F_i^d = f_{\text{mlp_}d}(F_i^d). \tag{12}$$

As same as the 2-D-Encoder, we obtain the fusion output $\hat{F}_i$ and add it to F_i as follows:

$$F_i = F_i + f_{fc}(\hat{F}_i). \tag{13}$$

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At last, aiming at modeling the global correlation between spectral and spatial features, we make use of multiscale attention network based on 3-D convolution layers to capture the spectral–spatial features in global range, which is more in line with the inherent structure of HSI. Concretely, we take F_i from (13) as the input and generate the attention map S as follows:

$$S = f_{\text{conv}}^{1\times1\times1}(F_i) + \sum_{r}^{\{1,2,5\}} f_{\text{conv}(r)}^{m\times5\times5}\left(F_i'\right) \tag{14}$$

where the kernel size on spectral dimension m is in the range $1, 1, 3$ with the dilation rate r in the range $1, 2, 5$ on height and weight.

As same as the 2-D-Encoder, we multiply the attention map and the input features as follows:

$$F_i = F_i + S \times F_i. \tag{15}$$

Based on the above operations, we can effectively obtain the spectral–spatial features that can provide enough discriminant information for label disambiguation.

B. Cross-Disambiguation Strategy

As mentioned above, label disambiguation plays a very important role in combating label ambiguity for PLL. Different from previous PLL methods that carry out label disambiguation solely within a single network, we perform cross-label disambiguation between two networks based on prototype learning [38], which can prevent the direct entanglement between feature representation and label disambiguation within the same network. Specifically, we maintain two sets of prototype embedding vectors $\mu^A = \{\mu_1^A, \mu_2^A, \ldots, \mu_K^A\}$ and $\mu^B = \{\mu_1^B, \mu_2^B, \ldots, \mu_K^B\}$ corresponding to each class for 2-D-Encoder and 3-D-Encoder respectively, which can be treated as the semantic representation of label space. As the clustering process, we transform the output features F_i', F_i of 2-D-Encoder and 3-D-Encoder into the same dimension of the label semantic representation q, k via MLP and compute the similarity between them to find the nearest label for updating the label confidence matrix $S^A = \{s_1^A, s_2^A, \ldots, s_n^A\}$, $S^B = \{s_1^B, s_2^B, \ldots, s_n^B\} \in [0, 1]^{n\times K}$, which is initially obtained by one-hot encoding and normalizing the candidate label set Y. s_i^A and s_i^B are the label confidence vectors of the ith sample.

1) Label Updating: We further propose a cross-update strategy based on moving-average mechanism. Specifically, we iteratively updating the label confidence matrix for each network according to the similarity between the output features and prototypes from another network, which can be expressed as follows:

$$\begin{cases} s_i^A = \gamma s_i^A + (1-\gamma) \times z^B \\ s_i^B = \gamma s_i^B + (1-\gamma) \times z^A \end{cases} \tag{16}$$

where γ is a constant between 0 and 1 (for convenience, we directly set γ to 0.99). And, $z^A = \{z_j^A\}_{j=1}^K and z^B = \{z_j^B\}_{j=1}^K$ are one-hot vectors of the pseudo label as follows:

$$z_c^A = \begin{cases} 1, & \text{if } c = \text{argmax}_{j\in Y}\ q^T \mu_j^A \\ 0, & \text{otherwise} \end{cases} \tag{17}$$

$$z_c^B = \begin{cases} 1, & \text{if } c = \text{argmax}_{j\in Y}\ k^T \mu_j^B \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

where μ_j^A and μ_j^B are the prototypes of 2-D-Encoder and 3-D-Encoder corresponding to the jth class, respectively. In this way, 2-D-Encoder and 3-D-Encoder can gradually update the label confidence matrix toward the correct one, ensuring a reliable and stable updating in a mutually supportive manner.

2) Prototype Updating: Similar to the iterative updating of cluster centers in clustering, the prototype vectors require synchronized updating in every training iteration. Then, we take the advantages of the normalized representation for samples with the pseudo-label j to update the prototype of class j, which is expressed as follows:

$$\begin{cases} \mu_j^A = \lambda \mu_j^A + (1-\lambda) q, & \text{if } q \in I_j^A \\ \mu_j^B = \lambda \mu_j^B + (1-\lambda) k, & \text{if } k \in I_j^B \end{cases} \tag{19}$$

where I_j^A and I_j^B denote the sample set with the pseudo-label j of 2-D-Encoder and 3-D-Encoder respectively, and λ is the updating weight between 0 and 1. For convenience, we directly set λ to 0.99.

C. Loss Function

1) Double-Consistency Loss Function: As mentioned above, we construct a heterogeneous network framework based on contrastive learning to achieve the highly effective and reliable semantic representation for label disambiguation. To achieve the feature complementarity across the two networks during the consistency constraint process, we establish constraints at both the feature and semantic levels. Specifically, we adopt the InfoNCE [39], [40] and the Wasserstein distance [41] to measure the consistency at the feature and semantic levels respectively, and the sum of them constitutes the consistency loss. The former has been widely used in representation learning and shows good performance, while the latter has been demonstrated the advantage of measuring the similarity of distributions even if there is no overlap between distributions or the overlap is negligible, so it is more friendly to model training. The details of feature consistency loss and semantic consistency loss are expressed as follows:

$$C_f = -\frac{1}{|D_b|} \sum_{q,k\in D_b} \log \frac{\exp\left(q^T k/\tau\right)}{\sum_{k'\in Q} \exp\left(q^T k'/\tau\right)} \tag{20}$$

$$C_s = W\left(p^A, p^B\right) = \inf_{\gamma\sim\pi\left(p^A, p^B\right)} E_{(x,y)\sim\gamma}\left[||x-y||\right] \tag{21}$$

$$\text{Loss}_c = \beta * C_f + C_s \tag{22}$$

where D_b and Q are the batch of samples and the queue of negative samples, respectively. q and k denote the semantic representations of the sample in D_b, and τ is the temperature. p^A and p^B are the predicted probability from the two networks (in testing stage, the final result is the weight sum of the predicted probability from the two networks.), and $W(\cdot)$ is the function of Wasserstein distance. Loss_c is the final consistency loss and β is a balanced parameter to control the feature consistency and the semantic consistency (for convenience, we directly set β to 0.5).

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Algorithm 1 Training for CrCD

Input: Training dataset D, 2-D-Encoder $N^A(\cdot)$, 3-D-Encoder $N^B(\cdot)$, MLP projector $M^A(\cdot)$ and $M^B(\cdot)$, classifier $cls^A(\cdot)$ and $cls^B(\cdot)$, batch size B, epochs T.

```
for t = 1, 2, ..., T do
    Split and shuffle D into |D|/B batches;
    for b = 1, 2, ..., |D|/B do
        Obtain the semantic representation for training
        samples by Eq.7, Eq. (15);
        Label prediction by cls^A(·), cls^B(·);
        Generate the prototypical similarity by Eq. (17);
        Update label confidence matrix by Eq. (16);
        Update prototypes by Eq. (19);
        Compute overall loss by Eq. (24)
        Update the parameters of overall network
    end for
end for
```

Output: parameters of the overall network.

2) Classification Loss Function: For multiclass classification, we make use of cross-entropy loss as the classification loss term, where $\text{Loss}^A_{\text{cls}}$ and $\text{Loss}^B_{\text{cls}}$ denote the classification loss of 2-D-Encoder and 3-D-Encoder, respectively. The loss function is defined as follows:

$$\text{Loss}^A_{\text{cls}} = -\frac{1}{|D_b|}\sum_{i=1}^{|D_b|}\sum_{j=1}^{K} s^A_{ij}\log\left(p^A_{ij}\right)$$
$$\text{Loss}^B_{\text{cls}} = -\frac{1}{|D_b|}\sum_{i=1}^{D_b}\sum_{j=1}^{K} s^B_{ij}\log\left(p^B_{ij}\right) \tag{23}$$

where s^A_{ij} and s^B_{ij} are the disambiguated label confidence of the jth label of the jth sample updated by (16), and p^A_{ij} and p^B_{ij} are the prediction probabilities of the jth label of the jth sample in the batch, and $|D_b|$ is the batch size.

For each network, its overall loss is the sum of the consistency loss and its classification loss as follows:

$$\text{Loss}^A = \text{Loss}^A_{\text{cls}} + \text{Loss}_c$$
$$\text{Loss}^B = \text{Loss}^B_{\text{cls}} + \text{Loss}_c. \tag{24}$$

We update two heterogeneous networks by stochastic gradient descent (SGD) separately. The pseudo-code is in Algorithm 1.

IV. Experimental Results and Discussion

In this section, we will verify CrCD on three hyperspectral classification datasets and compare with some of PLL algorithms.

A. Dataset Description

1) The Indian Pines dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 over a region with Indian Pine trees in Indiana, USA. The dataset has a size of 145 × 145 pixels with 224 spectral bands from 0.4 to 2.5μm. However, due to water reflection, the bands 104–108, 150–163, and 220 are typically excluded. Therefore, the focus of our study is usually on the remaining 200 bands after removing these 20 bands, which contains 16 types of land covers and its ground truth is shown in Fig. 5.

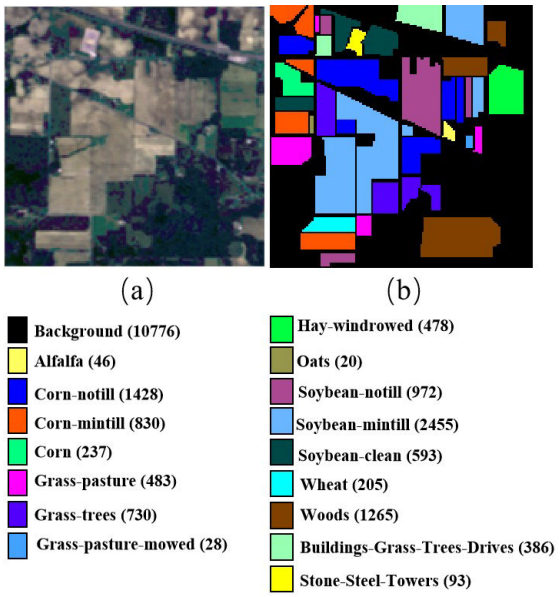


Fig. 5. Indian HSI. (a) False color. (b) Corresponding ground truth. (Note that the number of samples for each class is shown in brackets.)

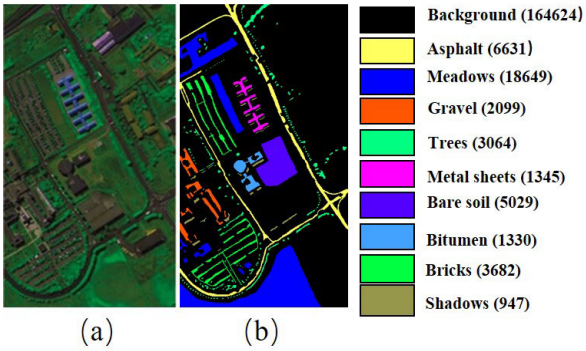


Fig. 6. PaviaU HSI. (a) False color. (b) Corresponding ground truth. (Note that the number of samples for each class is shown in brackets.)

2) The PaviaU dataset was captured over the city of Pavia, Italy, in 2003 by the Reflective Optics Spectrographic Imaging System (ROSIS-03), a German airborne reflectance optical spectroscopy imager. The spectroscopic imager continuously captured 115 bands in the wavelength range of 0.43–0.86 μm, with a spatial resolution of 1.3 m. Twelve bands were excluded due to noise, leaving 103 spectral bands typically for analysis. The data size is 610 × 340 pixels, which are shown in Fig. 6.

3) The HoustonU dataset was provided by the HSI analysis laboratory and the National Center for Airborne Laser Mapping (NCALM) at the University of Houston, which was captured by the ITRES-CASI 1500 hyperspectral imager that encompasses the University of Houston campus and its surrounding urban areas. The data has a size of 349 × 1905 pixels with 144 spectral bands from 380 to 1050 nm and the spatial resolution of 2.5 m, which includes 15 types of land covers and its ground truth is shown in Fig. 7.

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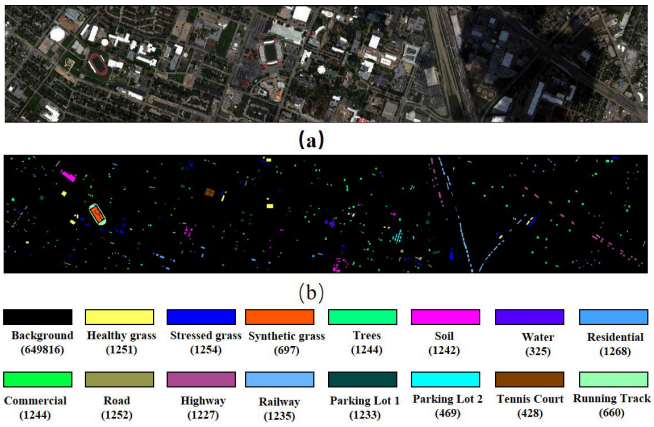


Fig. 7. HoustonU HSI. (a) False color. (b) Corresponding ground truth. (Note that the number of samples for each class is shown in brackets.)

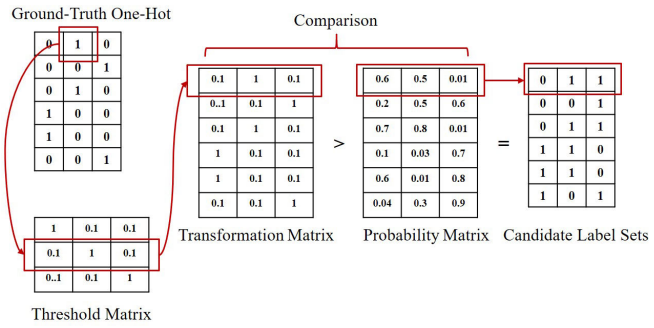


Fig. 8. Generation of the candidate label.

B. Comparison Models

To fully validate the advantages of CrCD, several PLL algorithms are utilized for comparison experiments, including two traditional machine learning based-methods, i.e., SLAP [24] and PL-AGGD [25], and three deep learning-based methods, i.e., PiCO [26], PaPi [32], and ParSE [31]. Additionally, three fully supervised HSI classification algorithms are also used for advanced studies, including one traditional machine learning-based method called LCMR [42], two deep learning-based methods called SSFTT [9] and Gnet [15], respectively.

C. Implementation

For all the datasets, we used three layers of multidirection MLP in 2-D-Encoder and 3-D-Encoder to explore the prototype features at long distance. We set two layers of MLP with ReLU activation layer for prototype learning. We utilized a single linear layer as classifier. We implemented the verification experiment of our method in the PyTorch environment, using an NVIDIA GeForce 3090Ti 24-GB GPU server. The SGD optimizer was selected as the initial optimizer, and the learning rate was set to 0.005. In batch training, we configured the size of each minibatch to be 64. Each dataset was randomly split into two groups: training set and test set, which has 5% of the total sample number in the training set and each category contains at least ten training samples. For each pixel, we set the patch size to 21×21.

As the same experimental settings in previous works [26], [32], we constructed partially labeled datasets by transforming negative labels $\overline{y} = y$ into false positive labels with a threshold $R = P(\overline{y} \in Y | \overline{y} \neq y)$. Specifically, as shown in Fig. 8, we set $R = 0.1$ and generated the threshold matrix for the label space with a value of 0.1 except for the diagonal line of 1. The diagonal line was used to locate the real labels and ensure that the real labels are among the candidate labels. The corresponding rows in the threshold matrix were then localized based on the true label in the one-hot matrix, which in turn resulted in the transformation matrix. Finally, the probability matrix to be false positive was randomly generated and compared with the threshold matrix to form the candidate set, if it is less than the threshold then it was considered as a candidate label and its value was set to 1, otherwise, it was a noncandidate label and set to 0. We set $R \in \{0.01, 0.05, 0.1\}$ for Indian Pines and HoustonU, and $R \in \{0.1, 0.2, 0.3\}$ for PaviaU, where the higher value of R is, the greater the ambiguity of labels is, and the more challenging the task of label disambiguation becomes.

D. Main Results

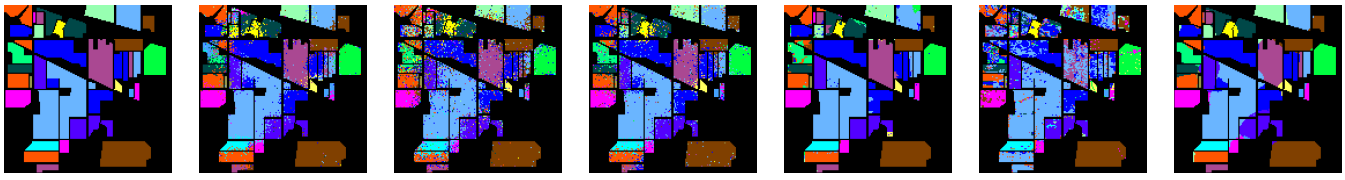
1) CrCD Achieves SOTA Results: First, as shown in Table I, we set $R = 0.1$ and utilize three quantitative evaluation indexes to validate the effectiveness of CrCD, including overall accuracy (OA), Kappa coefficient (KC), and the producer's accuracy (PA). The classification maps with $R = 0.1$ are shown in Figs. 9–11 on the Indian Pines, PaviaU, and HoustonU datasets, respectively. By combining Table I and Figs. 9–11 to observe, PL-AGGD shows the worst performance on all evaluation criteria, which is due to the fact that it only utilizes the spectral information and ignores the spatial information hidden in HSI; in this case, the adaptive graph-guided disambiguation strategy proposed by PL-AGGD may be not effective. By taking further observation, PiCO, PaPi, and ParSE achieve better performance than PL-AGGD but is still poor in identifying the types of land covers on the three datasets, and the main reason is that all of them directly utilize the ResNnet network based on CNN as backbone for feature extraction, which is good in common spatial feature extraction in the visual field of natural image and poor in spectral–spatial feature extraction on HSI. Although the maps of SLAP are clearer compared with the above three methods on the Indian Pines and PaviaU datasets, they separate representation learning and label disambiguation into two stages, which means that the performance ceiling of the latter is determined by the former. And it is worth noting that because the SLAP algorithm needs to build a superpixel-guided Laplacian matrix as prior for the subsequent process, it encounters difficulties when dealing with large-scale HSIs such as the HoustonU dataset. As a result, it is unable to provide experimental results for the SLAP algorithm on the HoustonU dataset, which is denoted as '–' in Table I. A synthesis of comparisons yields that the classification accuracy of CrCD is the best and the classification map is cleanest to the ground truth, which verifies the superior capability of CrCD in solving the PLL problem in HSI classification. The main reason is that

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TABLE I

CLASSIFICATION ACCURACY (OA, KC, PA) ON THE INDIAN PINES, PaviaU, HoustonU DATASETS WITH $R = 0.1$ (%)

Dataset	Class	PA					
		SLAP	PL-AGGD	PiCO	PaPi	ParSE	CrCD
Indian Pines	Alfalfa	**100.00**	61.11	97.22	83.33	100.00	**100.00**
	Corn-notill	88.19	59.22	84.01	63.23	84.08	**95.36**
	Corn-mintill	86.52	34.14	82.76	64.77	77.19	**94.93**
	Corn	79.20	15.11	19.03	57.52	82.30	**95.58**
	Grass-pasture	**94.96**	72.71	81.92	64.92	73.42	92.59
	Grass-Tree	**98.35**	91.20	84.44	71.04	80.98	94.09
	Grass-pasture-mowed	95.56	83.33	88.89	94.44	100.00	**100.00**
	Hay-windrowed	**100.00**	91.41	95.60	85.49	89.45	**100.00**
	Oats	100.00	60.00	90.00	70.00	60.00	**100.00**
	Soybean-notill	91.24	31.20	75.76	68.83	73.38	**91.56**
	Soybean-mintill	94.82	82.76	91.86	83.80	91.38	**95.37**
	Soybean-claen	84.69	32.68	78.90	49.47	72.52	**89.89**
	Wheat	99.48	91.75	71.28	73.33	78.46	**99.49**
	Woods	98.63	95.67	96.76	96.17	93.34	**99.00**
	Bulidings-Grass-Trees-Drives	91.69	20.49	72.21	66.21	74.39	**98.91**
	Stone-Steel-Towers	**94.94**	91.57	77.11	72.29	80.72	90.36
	AA	93.64	63.40	80.48	72.80	81.98	**96.07**
	OA	92.90	66.07	84.56	74.19	83.77	**95.30**
	KC	91.91	60.38	82.39	70.53	81.42	**94.65**
PaviaU	Asphalt	98.63	91.47	93.68	93.56	93.83	**99.17**
	Meadows	99.68	99.39	98.38	96.07	98.26	**99.90**
	Gravel	94.74	37.21	89.17	81.20	86.42	**98.45**
	Trees	90.84	87.25	78.01	72.28	78.70	**96.05**
	Matal sheets	97.27	99.53	91.94	82.63	90.06	**99.30**
	Bare soll	97.75	54.80	97.93	90.21	95.21	**99.79**
	Bitumen	97.32	5.15	95.17	92.41	94.54	**100.00**
	Bricks	97.91	93.97	95.00	88.77	90.59	**99.54**
	shadows	**98.35**	99.67	74.44	73.44	72.33	93.22
	AA	96.94	74.27	90.41	85.62	88.87	**98.38**
	OA	98.08	85.61	94.57	90.89	93.61	**99.23**
	KC	97.50	80.30	92.80	87.92	91.51	**98.98**
HoustonU	Healthy grass	–	**86.36**	85.79	87.47	84.19	86.29
	Stressed grass	–	83.96	75.92	65.52	70.39	**92.11**
	Synthetic grass	–	**100**	94.27	90.20	92.16	96.08
	Trees	–	**97.29**	85.79	77.92	77.33	90.02
	Soil	–	**99.75**	94.92	91.78	91.10	**99.75**
	Water	–	84.09	78.64	59.22	68.93	**86.41**
	Residential	–	90.61	84.32	73.61	76.18	**94.69**
	Commercial		57.66	69.04	65.57	65.06	**84.69**
	Road	–	85.87	81.18	70.34	74.12	**95.38**
	Highway	–	47.47	97.68	95.37	94.17	**99.40**
	Railway	–	43.90	87.65	83.90	86.71	**95.74**
	Parking Lot 1	–	18.70	86.95	83.19	79.61	**97.53**
	Parking Lot 2	–	2.92	79.15	82.96	81.61	**95.29**
	Tennis Court	–	90.89	98.03	95.58	92.63	**100.00**
	Running Track	–	99.04	84.21	90.43	89.79	**100.00**
	AA	–	72.57	85.57	80.87	81.59	**94.22**
	OA	–	72.56	85.41	80.55	81.03	**94.05**
	KC	–	70.28	84.16	78.96	79.50	**93.55**



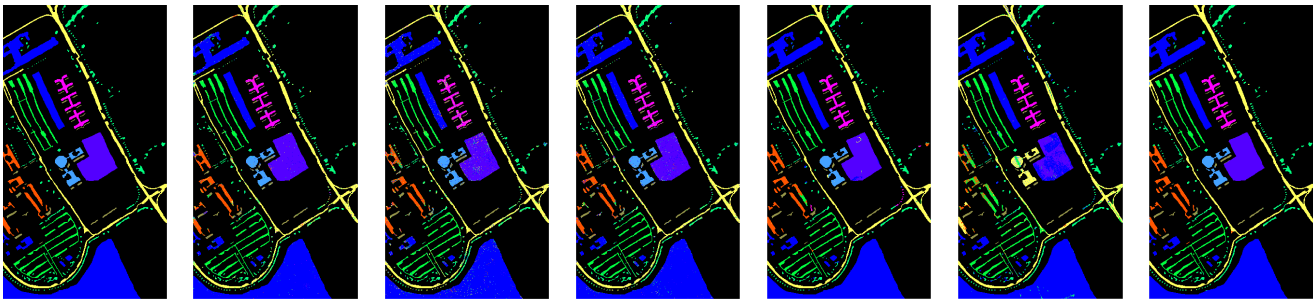
(a) (b) (c) (d) (e) (f) (g)

Fig. 9. Classification maps on the Indian Pines dataset with $R = 0.1$. (a) Ground truth. (b) PiCO. (c) PaPi. (d) ParSE. (e) SLAP. (f) PL-AGGD (g) CrCD.

we make full use of spectral–spatial information hidden in HSI by designing the heterogeneous network, and apply the more effective cross-disambiguation strategy for model training.

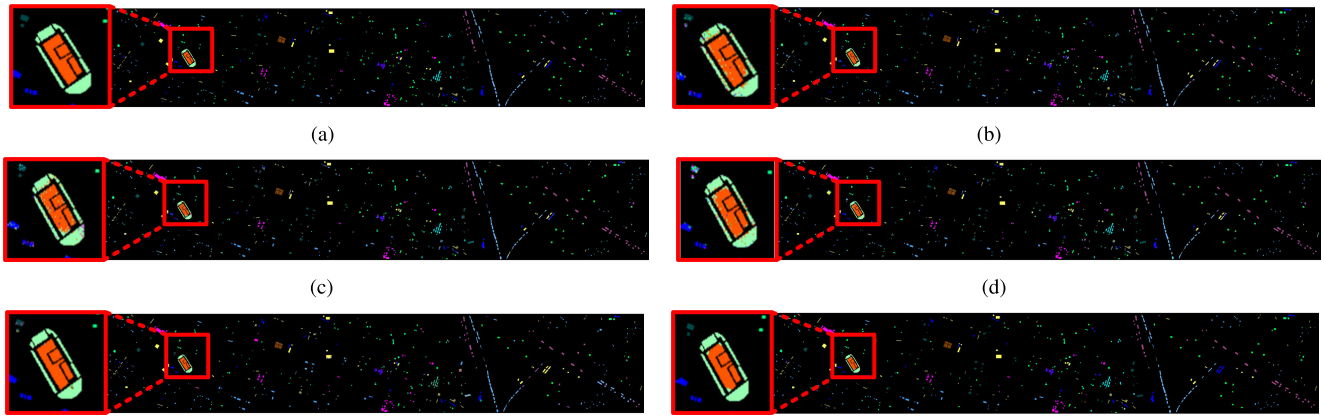
Additionally, the results under different values of R are presented in Tables II–IV on the Indian Pines, PaviaU, and HoustonU datasets, respectively. By taking insight into the classification performance of PL-AGGD, PiCO, PaPi, and ParSE that are not for HSI, it is obvious that the deep learning-based methods, i.e., PiCO, PaPi, and ParSE, can obtain better performance than the traditional machine learning-based method, i.e., PL-AGGD, which verifies the superiority of deep learning on HSI PLL. And then, it is worth to find that

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(a) (b) (c) (d) (e) (f) (g)

Fig. 10. Classification maps on the PaviaU dataset with $R = 0.1$. (a) Ground truth. (b) PiCO. (c) PaPi. (d) ParSE. (e) SLAP. (f) PL-AGGD. (g) CrCD.



(a) (b) (c) (d) (e) (f)

Fig. 11. Classification maps on the HoustonU dataset with $R = 0.1$. (a) Ground truth. (b) PiCO. (c) PaPi. (d) ParSE. (e) PL-AGGD (f) CrCD.

TABLE II

OAs WITH DIFFERENT R ON THE INDIAN PINES DATASET (%)

Dataset	Weakly Supervised			
	Models	R=0.01	R=0.05	R=0.1
Indian Pines	SLAP	94.64	93.41	92.90
	PL-AGGD	67.97	67.22	66.07
	PiCO	88.44	88.15	84.56
	PaPi	78.94	77.10	74.19
	ParSE	85.30	84.50	83.77
	CrCD (ours)	**96.78**	**96.35**	**95.30**
	Fully Supervised			
	LCMR		95.66	
	SSFTT		94.22	
	Gnet		95.07	
	CrCD(ours)		**96.86**	

TABLE III

OAs WITH DIFFERENT R ON THE PaviaU DATASET (%)

Dataset	Weakly Supervised			
	Models	R=0.1	R=0.2	R=0.3
PaviaU	SLAP	98.08	97.82	97.15
	PL-AGGD	85.61	85.08	84.47
	PiCO	94.57	94.67	94.37
	PaPi	90.89	89.25	86.55
	ParSE	93.61	93.00	92.36
	CrCD(ours)	**99.23**	**98.51**	**97.74**
	Fully Supervised			
	LCMR		99.31	
	SSFTT		98.86	
	Gnet		94.32	
	CrCD(ours)		**99.49**	

TABLE IV

OAs WITH DIFFERENT R ON THE HoustonU DATASET (%)

Dataset	Weakly supervised methods			
	Models	R=0.01	R=0.05	R=0.1
HoustonU	PL-AGGD	75.28	74.22	72.56
	PiCO	87.47	85.98	85.41
	PaPi	83.04	82.40	80.55
	ParSE	82.09	80.96	81.03
	CrCD(ours)	**96.06**	**94.99**	**94.05**
	Fully supervised methods			
	LCMR		93.09	
	SSFTT		95.41	
	Gnet		90.31	
	CrCD(ours)		**96.25**	

the traditional machine learning-based method (i.e., SLAP) outperforms the above methods with various partial label thresholds. The reason is that the above methods from the other tasks do not combine the PLL and the characteristics of HSIs.Moreover, it is obvious that the proposed CrCD shows significant superiority over all the comparison methods on all the datasets with various partial label thresholds, particularly excelling in scenarios with high levels of ambiguity, which is due to the fact that CrCD adequately applies the advantages of deep learning to the HSI PLL. Specifically, CrCD outperforms the current models by more than 2% and 8% on the Indian Pines and HoustonU datasets when R is set from 0.01 to 0.1 respectively. On the PaviaU dataset, CrCD is still superior to SLAP by 0.59% when R is set to 0.3. Most significantly, CrCD achieves comparable results against the model under full supervision, which demonstrates the satisfactory label disambiguation of CrCD. Taking the results in Table II as

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TABLE V
OAs FOR ABLATION STUDY ON THE INDIAN PINES, PaviaU, AND HoustonU DATASETS (%)

Ablation	Signal-Direction MLP	Multi-Direction MLP	Multi-Scale Attention	C_s	Indian Pines ($R = 0.1$)	PaviaU ($R = 0.1$)	HoustonU ($R = 0.1$)
Feature Extraction	×	×	×	×	91.60	98.15	88.18
	×	✓	×	×	91.61	98.51	89.37
	×	✓	✓	×	93.26	99.08	92.79
Label Disambiguation	×	✓	✓	×	95.25	99.18	93.89
	✓	×	✓	✓	92.23	99.39	93.82
	×	✓	✓	✓	95.30	99.23	94.05

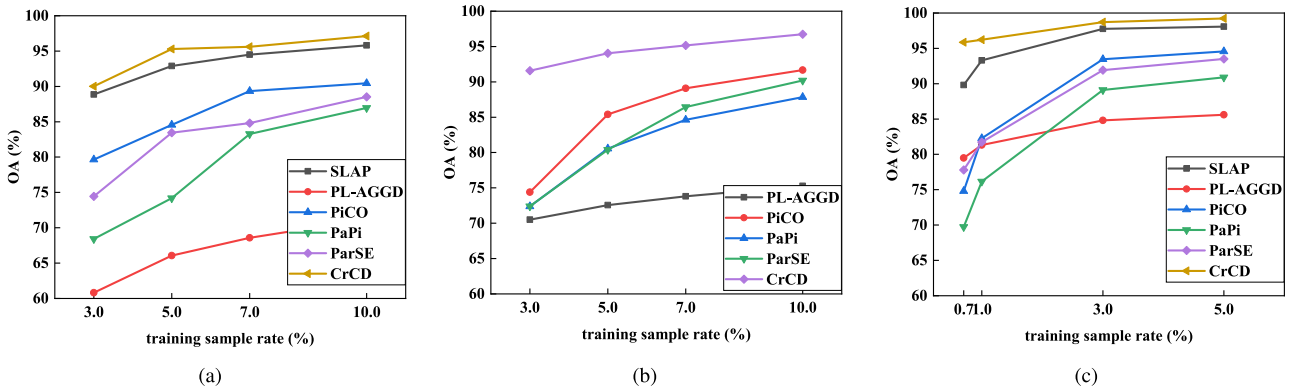


Fig. 12. Classification accuracy with respect to different ratios of training samples. (a) Indian Pines. (b) HoustonU. (c) PaviaU.

example, the difference between the weakly supervised model and the supervised model is less than 1.6% in different cases. In Table III, the performance of CrCD degrades by less than 2% compared with the model under full supervision when R is set to 0.3.

Last but not the least, we further validate the model performance under the fully supervised conditions by comparing with two supervised methods. It is not difficult to observe that CrCD demonstrates an improvement of at least 1% compared with SSFTT and GNet, showing the superiority of CrCD in feature extraction and classification. For example, CrCD outperforms LCMR by 1.2% and 0.18% on the Indian Pines and PaviaU datasets in Table II and Tabel III, SSFTT by 0.84% on the HoustonU dataset in Table IV respectively, which demonstrates the superior feature extraction capability of our model.

From the comprehensive analysis of the results in Tables II–IV, I, and Figs. 9–11, it can be observed that our model consistently outperforms all the comparison methods. All in all, the entire comparison results underscore that CrCD has the ability and superiority in extracting discriminative features and identifying the ground truth.

E. Ablation Studies

To fully validate the effectiveness of CrCD in this article, the ablation experiments were conducted according to the combination of different components on the three datasets and analyzed in terms of OA. All the experimental results are presented in Table V.

1) Effectiveness of Feature Extraction: For representation learning, under the condition of no label disambiguation,

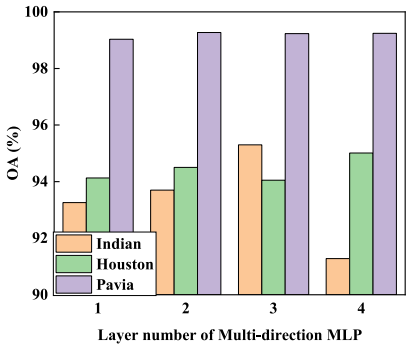


Fig. 13. Classification accuracy with respect to different layer numbers of multidirection MLP (training samples: 5%).

we validate the contribution of each component for feature extraction: multidirection MLP and multiscale attention. As shown in Table V, it is notable to observe that the classification accuracies on all the datasets gradually improve with the stacking of module, which thoroughly demonstrates the effectiveness of each component. For example, the baseline model without the proposed modules under the feature consistency loss C_f yielded the worst classification accuracy. By adding the multidirection MLP to the first case, the model performed better over more than 1% on the HoustonU dataset. By adding the multiscale attention to the second case, the model performed better over more than 3% on the HoustonU dataset. Additionally, to validate the effectiveness of multidirection MLP, we compare it with a single-direction MLP that just considers the dimensional information and presents the results in Table V. Above all, this indicates that the CrCD has stronger ability in representation learning, which can provide discriminative information for label identification.

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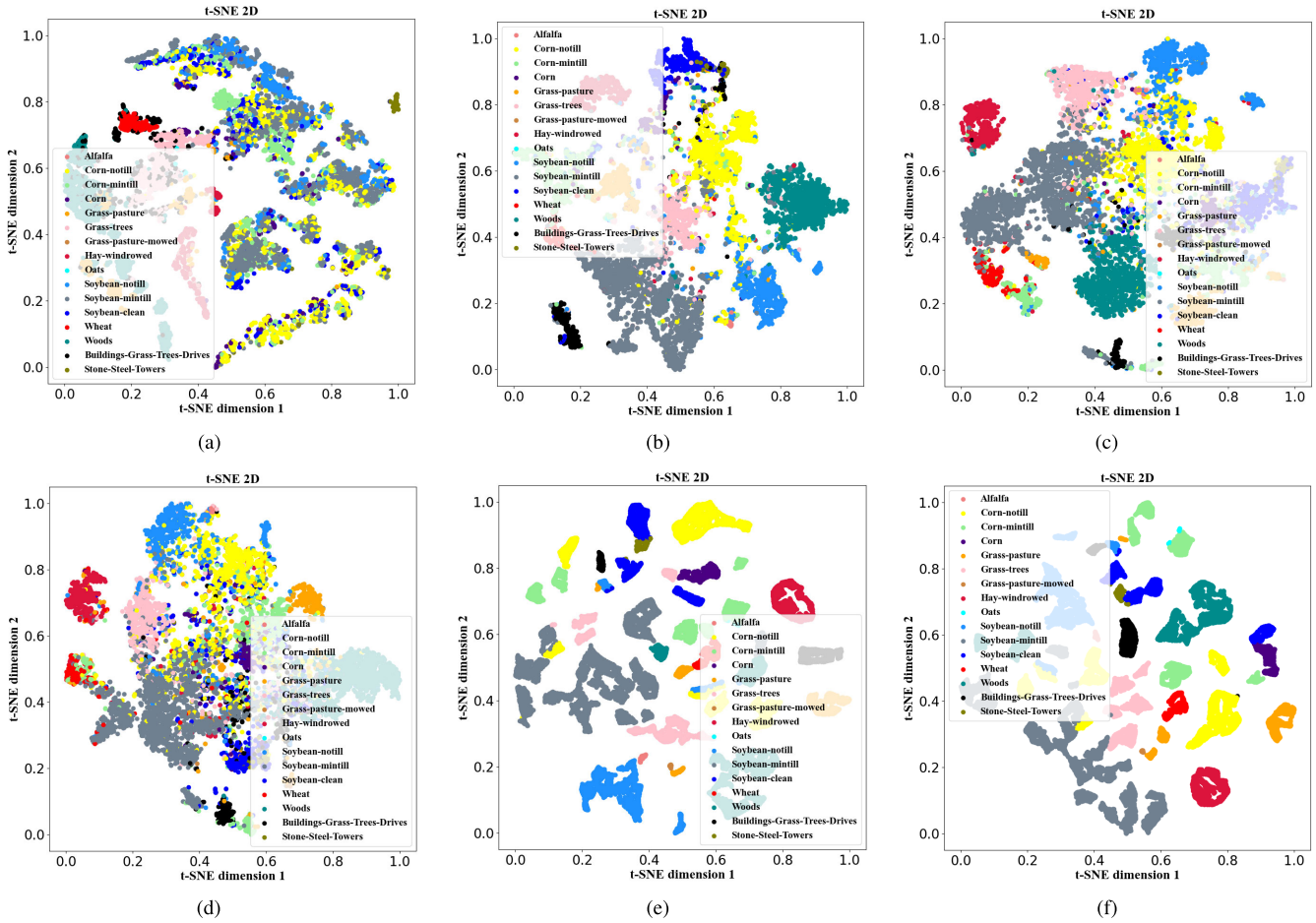


Fig. 14. Representation visualization with t-SNE on the Indian Pines dataset with $R = 0.1$. (a) Raw. (b) PiCO. (c) ParSE. (d) PaPi. (e) Only 2-D-Encoder in CrCD. (f) Only 3-D-Encoder in CrCD.

2) Effectiveness of C_s and Label Disambiguation: Based on the above representation learning method, we further report the classification accuracies for the cross-label disambiguation strategy in Table V. It is obvious that the strategy of cross-label disambiguation can significantly enhance classification accuracy. In other words, it can effectively eliminate label ambiguity. Specifically, the model with cross-label disambiguation outperforms the best module without it. Besides, we can observe that the model with the constraint on semantic prediction achieves better performance than the one without it, which signifies the significance of semantic consistency for contrastive learning.

F. Parameter and Visual Analysis

1) Parameter Analysis: In the parameter analysis, we analyzed the influence of the training sample ratio and the layer number of multidirection MLP.

Fig. 12 shows the impact of different ratio of training samples on the classification accuracies of the Indian Pines, PaviaU, and HoustonU datasets when R is set to 0.1. The training sample ratios are in the range of [3%, 5%, 7%, 10%] on the Indian Pines and HoustonU datasets, and [0.7%, 1%, 3%, 5%] on the PaviaU dataset. It can be seen from Fig. 12 that the classification accuracy is positively correlated with the proportion of training samples, meaning that as the proportion of training samples increases, the classification accuracy also improves. The reason is that the model can learn more information for classification when the proportion of training samples increases. Moreover, it is worthy to note that as the number of training samples increases, the rate of improvement in accuracy gradually diminishes. The reason may be information saturation, namely the current information is already sufficient, and there is no new information in the surplus data. Last but not the least, CrCD outperforms all the comparison methods under different training sample ratios, especially for the low training sample ratios, which fully demonstrates its robustness on classification accuracy when facing the changing on the number of training samples.

Fig. 13 shows the influence of different layer numbers of multidirection MLP on the classification accuracy for the Indian Pines, PaviaU, and HoustonU datasets. The layer number of multidirection MLP is in the range of [1–4], R is set to 0.1, and the training sample ratio is set to 5%. It can be observed from Fig. 13 that the classification accuracies tend to increase to a certain extent with the addition of network layers. However, once it reaches a peak, further increasing the number of layers will introduce additional noise, subsequently leading to a decline in classification performance. Specifically, the optimal performance is achieved on the Indian Pines

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dataset when the network has three layers, but there is a decline in accuracy when the number of layers is increased to 4. Additionally, it is noteworthy that the optimal number of network layers varies across different datasets, indicating the necessity for appropriate adjustments according to the characteristics of datasets.

2) Visual Analysis: To verify whether CrCD is effective to obtain distinguishable representation, we visualize the representation of PiCO, ParSE, PaPi, and CrCD using t-SNE in Fig. 14. All of them are the deep-learning-based methods. We can find out that the feature representation of CrCD shows more remarkable intraclass compactness and clearer interclass boundaries, which serves as validation for its efficacy in acquiring high-quality representation. In summary, the extensive experiments demonstrate the superiority of CrCD in both feature representation and classification accuracy, which effectively alleviates the entangled challenges of representation learning and label disambiguation.

V. Conclusion

In this article, we propose a label disambiguation model, termed multidirection-MLP-based CrCD for HSI PLL. In this model, we design a coherent framework to solve the key problem of the mutual entanglement between representation learning and label disambiguation in HSI PLL. Specifically, there are two heterogeneous networks framework based on contrastive learning, which can provide high-quality representation for label disambiguation. Based on the representation from the network, we further design cross-label disambiguation strategy to provide reliable supervision information for feature learning. The interplay between these two components mutually reinforces, resulting in a high-performance HSI classifier, which can accurately identify the ground-truth types of land covers. CrCD achieves the state-of-art performance on the Indian Pines, PaviaU, and HoustonU datasets in HSI PPL compared with the other related methods. In the future work, we will further explore and address the challenges in HSI classification within the realm of weak supervision, which will alleviate the pressure of labeling and reduce the associated costs in turn.

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